

# 1 equations

ALL IN THIS 1-2:

$$\frac{A_2}{B_2} = 8\pi\bar{h}\frac{\mu^3}{\lambda^3};$$

$$g_1B_{12} = g_2B_{21}$$

$$-dn_2(\nu) = A_{21}n_2f(\nu)d\nu;$$

$$-dn_2 = B_{21}n_2f(\nu)\rho_\nu d\nu$$

$$f(\nu) = \frac{\frac{\Delta\nu}{2\pi}}{(\nu-\nu_0)^2+(\frac{\Delta\nu}{2})^2};$$

$$\Delta\nu = \frac{1}{2\pi\tau};$$

$$\text{GAP} = \sqrt{1 + \frac{I}{I_s}\Delta\nu};$$

$$\text{Stable } \textbf{Condition}: 0 < (1 - \frac{L}{R_1})(1 - \frac{L}{R_2}) < 1;$$

$$\Delta n = \frac{[(\nu-\nu_0)^2+(\frac{\Delta\nu}{2})^2]\Delta n_0}{(\nu-\nu_0)^2+(1+\frac{I}{I_s})(\frac{\Delta\nu}{2})^2} = \frac{\Delta n_0}{1+\frac{I}{I_s}\frac{f(\nu)}{f(\nu_0)}};$$

$$\Delta n = n_2 - \frac{g_2}{g_1}n_1;$$

$$G = \Delta n B_{21} f(\nu) h \nu \frac{m u}{c};$$

$$\Delta n = R_2 \tau_2 - (R_1 + R_2) \tau_1;$$

$$G = \frac{[(\nu-\nu_0)^2+(\frac{\Delta\nu}{2})^2]G^0}{(\nu-\nu_0)^2+(1+\frac{I}{I_s})(\frac{\Delta\nu}{2})^2} = \frac{G^0}{1+\frac{I}{I_s}\frac{f(\nu)}{f(\nu_0)}};$$

$$G^0(\nu) = \Delta n_0 B_{21} \frac{\mu}{c} f(\nu) h \nu_0$$

$$\Delta n(\nu) = \frac{[(\nu-\nu_0)^2+(\frac{\Delta\nu}{2})^2]\Delta n_0 f_D(\nu)}{(\nu-\nu_0)^2+(1+\frac{I}{I_s})(\frac{\Delta\nu}{2})^2} = \frac{\Delta n_0 f_D(\nu)}{1+\frac{I}{I_s}\frac{f(\nu)}{f(\nu_0)}};$$

$$\text{wide of hole } \nu \pm \sqrt{1 + \frac{I}{I_s} \frac{\Delta\nu}{2}}$$

$$G_D(\nu_1) = \frac{G_d^0(\nu_1)}{\sqrt{1+\frac{I}{I_s}}}$$

$$a_{all} = a_{in} - \frac{1}{2L} \ln r_1 r_2 = -\frac{1}{2L} \ln(1 - (a_1 + t_1))$$

$$\textbf{Condition}: 2\delta\Phi = 2q\pi, q = 1, 2, 3...$$

$$\Delta\nu_q = \frac{c}{2\mu L}$$

$$2L = q\lambda$$

$$\text{square} \Delta\Phi_{mn} = (m+n+1)\frac{\pi}{2}$$

$$\nu_{nmq} = \frac{c}{2\mu L}(q + \frac{1}{2}(m+n+1))$$

$$\frac{1}{2}\Delta\nu_q = \Delta\nu_m = \Delta\nu_n$$

$$\text{round} \Delta\Phi_{mn} = (m+2n+1)\frac{\pi}{2}$$

$$\nu_{nmq} = \frac{c}{2\mu L}(q + \frac{1}{2}(m+2n+1))$$

$$\frac{1}{2}\Delta\nu_q = \Delta\nu_m = \frac{1}{2}\Delta\nu_n$$

$$\textbf{Gauss:}$$

$$\omega(z) = \omega_0 \sqrt{1 + (\frac{f}{z})^2}$$

$$\omega_0 = \sqrt{\frac{\lambda f}{\pi}}$$

$$R(z) = z[1 + (\frac{f}{z})^2]$$

$$L = |z_1| + |z_2|$$

$$R_1 = R_2 = 2f(\text{S.C chamber}) \text{ not suitable for normal shpere.}$$

$$f = \frac{\pi\omega_0^2}{\lambda}$$

$$2\theta = \frac{2\lambda}{\pi\omega_0}$$

$$\begin{aligned}
q &= z + if \\
2\theta_{mn} &= \sqrt{2m+1}\sqrt{2n+1}2\theta_0 \\
V_{00}^0 &= \frac{L^2\lambda}{2} \\
V_{mn}^0 &= \sqrt{2m+1}\sqrt{2n+1}V_{00}^0 \\
&\text{HOMOS} \\
G &= \frac{G^0}{1+2\frac{I}{I_s}} = G_{Thres} \\
a_{all} &= \frac{a_1+t_1}{2L} \\
I &= \frac{I_s}{2}t_1\left(\frac{2LG^0}{a_1+t_1} - 1\right) \\
P &= IS \\
&\text{non-HOMOS} \\
\text{non-center franquency :} G(\nu) &= \frac{G_D^0(\nu)}{sqr{t1+\frac{I}{I_s}}} = G_{Thres} \\
\text{center franquency :} G(\nu) &= \frac{G_D^0(\nu)}{sqr{t1+2\frac{I}{I_s}}} = G_{Thres}
\end{aligned}$$

## 2 homeworks

all:

chapter1: 2,3,4,11,12,13 P28

chapter2: 1,2,3,7,8,10,11,12,13 P48

chapter3: 1,2,4,6,7,9,10 P91

chapter4: 1,2,3,4,5,8,9 P119

### 3 keys in chapter 1

CHAPTER1 Formulas used in homework.

$$n \sim g e^{-\frac{E}{kT}} \quad (1)$$

$$\frac{q_s}{q_{normal}} = \frac{c^3}{8\pi h \nu^3} = \frac{1}{e^{\frac{h\nu}{kT}} - 1} \quad (2)$$

$$I = I_0 e^{-Gz} \quad (3)$$

$$\nu = \nu_0 \left(1 + \frac{v}{c}\right) \quad (4)$$

$$f(\nu) = \frac{\frac{\Delta\nu}{2\pi}}{(\nu - \nu_0)^2 + \left(\frac{\Delta\nu}{2}\right)^2} \quad \Delta\nu = \frac{1}{2\pi\tau} \quad (5)$$

$$\rho(\nu) = \frac{\frac{8\pi h \nu^3}{c^3}}{e^{\frac{h\nu}{kT}} - 1} \quad (6)$$

$$(7)$$

**SOME PROPOSITION NEEDED TO BE PROVED IN CHAPTER 1:**

I:PROVE:

$$A = \frac{1}{\tau} \quad (8)$$

prove:

$$\begin{aligned} n_2(t) &= n_{20} e^{At} \\ \text{consider } n : n &= n_0 e^{\frac{t}{\tau}} \\ \text{so : } A_{21} &= \frac{1}{\tau} \end{aligned}$$

II:PROVE:

$$A_{21} = B_{21} \frac{8\pi h \nu^3}{c^3} \quad (9)$$

$$B_{21} = B_{12} \frac{g_1}{g_2} \quad (10)$$

prove:

$$\begin{aligned} A_{21} n_2 dt + B_{21} n_2 \rho_\nu dt &= B_{12} n_1 \rho_\nu dt \\ (A_{21} + B_{21} \rho_\nu) \frac{n_2}{n_1} &= B_{12} \rho_\nu \end{aligned}$$

consider plank  $\rho_\nu$  :

$$n \sim g e^{\frac{-E}{kT}}$$

$$\frac{n_2}{n_1} = \frac{g_2}{g_1} e^{-\frac{h\nu}{kT}}$$

$$A_{21} = \left( \frac{g_1}{g_2} e^{\frac{h\nu}{kT}} B_{12} - B_{21} \right) \rho_\nu$$

$$\rho_\nu = \frac{A_{21}}{\frac{g_1}{g_2} e^{\frac{h\nu}{kT}} B_{12} - B_{21}}$$

$$\rho_\nu = \frac{A_{21} \frac{g_2}{g_1 B_{12}}}{e^{\frac{h\nu}{kT}} - \frac{g_2}{g_1 B_{12}} B_{21}}$$

$$\text{compare with : } \rho_\nu = \frac{\frac{8\pi h\nu^3}{c^3}}{e^{\frac{h\nu}{kT}} - 1}$$

so we have:

$$\frac{8\pi h\nu^3}{c^3} = A_{21} \frac{g_2}{g_1 B_{12}} \quad (11)$$

$$1 = \frac{B_{21} g_2}{B_{12} g_1} \quad (12)$$

## 4 keys in chapter 2

SOME KEYS IN HOMEWORK:

$$0 < g_1 g_2 < 1; g_1 = 1 - \frac{L}{R_1}, g_2 = 1 - \frac{L}{R_2}$$

$$G = \Delta h \nu \frac{\mu}{c} f(\nu) B_{21}$$

$$\Delta n_0 = R_2 \tau_2 - (R_1 + R_2) \tau_1$$

$$\Delta n = \frac{\Delta n_0}{1 + \frac{I f(\nu)}{I_s f(\nu_0)}}$$

$$G = \frac{G_0(\nu)}{1 + \frac{I f(\nu)}{I_s f(\nu_0)}} = \frac{(\frac{\Delta\nu}{2})^2 G_0(\nu_0)}{(\nu - \nu_0)^2 + (1 + \frac{I}{I_s})(\frac{\Delta\nu}{2})^2} \quad \Delta\nu' = \sqrt{1 + \frac{I}{I_s}} \Delta\nu$$

$$G_D = \frac{G_0(\nu) f_D(\nu)}{1 + \frac{I f(\nu)}{I_s f(\nu_0)}}$$

$$G_D(\nu_1) = \frac{G_D^0(\nu_1)}{(1 + \frac{I}{I_s})^{\frac{1}{2}}}$$

$$a_{whole} = a_{iner} - \frac{1}{2L} \ln(r_1 r_2)$$

$$T_L = \begin{bmatrix} 1 & L \\ 0 & 1 \end{bmatrix}; T_f = \begin{bmatrix} 1 & 0 \\ -\frac{1}{f} & 1 \end{bmatrix}; T_{\frac{n_1}{n_2}} = \begin{bmatrix} 1 & 0 \\ 0 & \frac{n_1}{n_2} \end{bmatrix}$$

## 5 keys in chapter 3

$$R = z + \frac{f}{z}$$

$$\omega = \omega_0 \sqrt{1 + \left(\frac{z}{f}\right)^2}$$

$$\theta_0 \omega_0 = \frac{\lambda}{\pi}$$

$$\omega_0 = \sqrt{\frac{\lambda f}{\pi}}$$

$$q = z + if$$

$$M^2 = \frac{\omega \theta}{\omega_0 \theta_0}$$

$$P_o = \frac{1}{2} t_1 I_s A \left( \frac{2LG_0}{a_1 + t_1} - 1 \right)$$

$$P_e(\nu_1) = t_1 I_s A \left[ \left( \frac{2LG_D^0}{a_1 + t_1} \right)^2 - 1 \right]; P_e(\nu_0) = \frac{1}{2} I_s A t_1 \left[ \left( \frac{2LG_D^0(\nu_0)}{a_1 + t_1} \right) - 1 \right]$$

$$\text{best } t : \frac{dP}{dt_1} = 0$$

$$\text{square: } \phi_{mn} = \frac{c}{2\mu L} \left( q + \frac{1}{2}(m+n+1) \right)$$

$$\text{round: } \phi_{mn} = \frac{c}{2\mu L} \left( q + (m+2n+1) \right)$$

$$\text{LG:like square } \omega_{mn} = \sqrt{m+n+1} \omega_{00}$$

$$\text{LG: } \theta_{00} = \sqrt{m+n+1} \theta_{00}$$

$$\text{HG:Hermit Gauss : } \omega_m = \sqrt{2m+1} \omega_{00}; \omega_n = \sqrt{2n+1} \omega_{00}$$

$$\text{HG:Hermit Gauss : } \theta_m = \sqrt{2m+1} \theta_{00}; \theta_n = \sqrt{2n+1} \theta_{00}$$

## 6 keys in chapter 4

$$\tau = \frac{T}{2N+1}$$

$$P_{locked} = NP_{aver} = N^2 P_{single}$$

$$M = \frac{f_2 \omega}{f_0 \omega_0} \text{ f is the focus distance } \omega \text{ is the r at Mirror}$$

NOTICE: the  $q = z + i\Gamma$ , the  $z$  is distance from  $\omega_0$  so can just don't write the transport matrix form  $\omega_0$  to the lens